

CORN CROP DISEASE CLASSIFICATION

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INTRODUCTION

In this document, our group will present a focused analysis of the rationale, objectives, and approach behind developing our corn crop disease classification project. The purpose of this project is to provide Ontario farmers with an early disease detection tool for corn crops, targeting Common Rust, Gray Leaf Spot (GLS), and Northern Corn Leaf Blight (NLB). Crop disease is a leading cause of yield loss and can severely impact both a farmer's livelihood and Ontario's agricultural economy. Currently, farmers must either rely on costly and time-consuming laboratory testing or on visual inspection, which can be unreliable since symptoms are often subtle, appear late, and may be misidentified. Deep learning, particularly convolutional neural networks, is well-suited to offer a third, more convenient option that is neither costly, time-consuming, or prone to inaccuracies. These models can identify faint visual patterns beyond human perception, deliver consistent and fatigue-free analysis, and scale efficiently to large and diverse datasets. Our project takes advantage of these strengths by applying this technology to real-world corn leaf imagery, providing farmers with a reliable and accessible tool for early, cost-effective, and accurate detection that works reliably in real-world field conditions and is easy for non-expert users to adopt and operate. See Figure 1.

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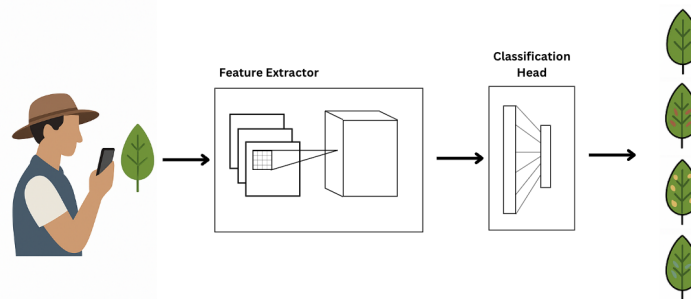


Figure 1: Simplified overview of the use of our model

1 BACKGROUND & RELATED WORK

Given the economic and agricultural importance of corn in Ontario, early and accurate disease detection is critical. Traditional methods rely on manual inspection, which is time-consuming, labour-intensive, and often requires extensive infrastructure. Recently, researchers have explored the po-

tential of deep learning to address this challenge. CNNs, in particular, have emerged as a powerful tool in plant disease classification due to their ability to extract features from images.

Early work by Mohanty et al. (2016) applied CNNs like AlexNet and GoogLeNet to 54,306 lab-acquired leaf images, achieving 99.35% accuracy across 26 classes, including our targeted corn diseases. While groundbreaking, the dataset’s clean backgrounds limited applicability to real-world field conditions.

Ferentinos (2018) expanded on this with 87,000 images, 37.3% from field conditions, achieving 99.53% accuracy across 58 crop-disease combinations. This demonstrated stronger generalization but still relied heavily on lab-curated imagery.

To bridge the gap between lab and field conditions, Ahmad et al. (2021) introduced the CD&S dataset, containing 2,112 raw field images of four corn categories, plus 2,343 augmented samples and bounding box annotations. This dataset supports both classification and detection tasks in realistic agricultural settings.

More recently, Li et al. (2024) optimized YOLOv8s with a GhostNet backbone and triplet attention, achieving 87.5% precision with only 11.2 MB model size, ideal for real-time, low-resource deployment.

Finally, Mohanty et al. (2023) compared CNN architectures on 4,800 corn leaf images (rust, gray leaf spot, blight, healthy). DenseNet121 and VGG19 achieved above 99% accuracy, while a ResNet50-VGG16 hybrid reached 99.65%, reinforcing CNNs’ effectiveness for corn disease classification.

2 DATA PROCESSING

This section describes the methods used to prepare data for our deep learning model and test its performance. We outline the data sources, steps for cleaning and formatting images, and how we structured the dataset into training, validation, and test splits. We also provide statistics and examples of the cleaned dataset, and explain our plan for testing on entirely new data.

2.1 DATASETS OVERVIEW

The first dataset is **PlantDoc** (Singh et al., 2020), containing over 2,500 images across 27 classes from 13 plant species. Images are captured in real-world conditions with varied backgrounds, inconsistent lighting, and occlusions. To keep corn unseen for evaluation, we **exclude all corn images**. PlantDoc’s variability makes it well-suited for learning robust visual features.

The second dataset, **PlantVillage** (Hughes & Salathé, 2015), has about 54,000 images across 39 classes, captured in controlled lab settings with clean backgrounds and consistent lighting. See Figure 2. We also **exclude all corn images** to prevent leakage and ensure later corn performance reflects true generalization. PlantVillage complements PlantDoc by providing cleaner, higher-quality samples.



Figure 2: PlantVillage



Figure 3: CD&S

The third dataset, **Corn Diseases & Symptoms (CD&S)** (Ahmad et al., 2021), contains 2,112 expert-labeled corn leaf images in four classes, of which we use Healthy, Gray Leaf Spot, and Northern Corn Leaf Blight. All are collected under real-world field conditions. See Figure 3.

The fourth dataset, **Roboflow Corn Diseases** (7, 2022), has 3,647 labeled corn images in Common Rust (1,177), Gray Leaf Spot (1,161), and Leaf Blight (1,309), matching our disease targets. Images vary in lighting, orientation, and background clutter.

The fifth dataset, **IDADP** (Lei et al., 2025), has 3,600 images evenly split into six classes; we use only Common Rust, Gray Leaf Spot, and Northern Corn Leaf Blight.

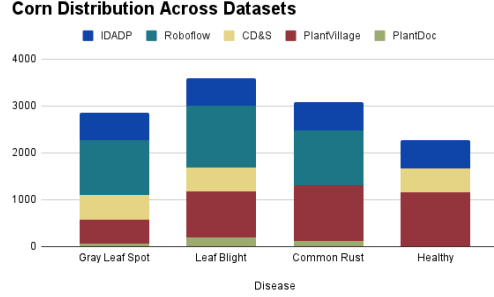


Figure 4: Class distribution among datasets.

In summary, **PlantDoc** and **PlantVillage** are used to pretrain the feature extractor, with corn removed to preserve it as unseen during this stage. The **CD&S** dataset, plus the excluded corn images from PlantDoc and PlantVillage, train and validate the classification head. **Roboflow** and **IDADP** are kept strictly for final testing to assess generalization to new sources and conditions. See Figure 4 to see how each dataset contributes to each class.

2.2 DATA CLEANING AND FORMATTING

For **PlantDoc**, we created a function to scan all image files, store attributes (file paths, class names, labels), and save them in a pickle file for faster loading.

All datasets follow the `ImageFolder` format, with each class in its own folder. In **PlantDoc** and **PlantVillage**, folders use the format `plant__disease`, so we split names at the double underscore to separate plant and disease. We also built a disease–plant dictionary for filtering.

To ensure **LargeNet** pretraining used only non-corn images, we removed all corn classes from PlantDoc and PlantVillage by skipping any folder starting with `corn`. This avoided corn exposure during pretraining and prevented data leakage.

The **CD&S** dataset was already clean and organized, with no corrupted files.

Overall, this process ensured consistent structure, removed unwanted classes, and streamlined loading.

2.3 DATA AUGMENTATION AND PREPROCESSING

We applied data augmentation to simulate real-world variation in lighting, orientation, and image quality, improving model robustness. Transformations used `torchvision.transforms` in `get_cached_data_loaders()`.

We defined three augmentation levels via the `augmentation_strength` argument:

- **Light:** Resize to 256×256 , random crop to 224×224 , horizontal flip (50%), normalization.
- **Medium:** Light augmentations plus rotation (up to 15°) and brightness/contrast jitter.
- **Heavy:** Medium augmentations plus stronger jitter, vertical flip (30%), and rotation (up to 30°).

All images are resized to 224×224 to match CNN input requirements. These augmentations increased sample diversity without extra data collection.

2.4 TRAINING AND VALIDATION SPLIT

Our **LargeNet** pipeline begins with feature extractor pretraining on **PlantDoc** and **PlantVillage** (corn excluded) to learn plant-agnostic disease features. The classification head (ANN) is trained on the **CD&S** dataset plus excluded corn images from PlantDoc and PlantVillage, covering our four target classes.

We use a 70/15/15 split for training, validation, and internal testing, with `torch.manual_seed(42)` to ensure reproducibility. Class balance is maintained, and DataLoaders shuffle training data while keeping validation and test sets fixed.

Roboflow and **IDADP** are used only for final evaluation, remaining unseen during both pretraining and classification head training, to measure generalization to new, real-world data distributions.

3 ARCHITECTURE

We use a single convolutional architecture, **LargeNet**, an AlexNet-inspired network enhanced with several modern deep learning practices to improve performance and generalization.

3.1 STRUCTURE AND PRETRAINING

LargeNet consists of:

- **Five convolutional blocks** with channel depths: $64 \rightarrow 192 \rightarrow 384 \rightarrow 256 \rightarrow 256$. The increase to 384 channels allows the network to capture complex spatial features, while the subsequent reduction to 256 compresses and removes redundancy.
- **Batch Normalization** within each block to stabilize and speed up training.
- **ReLU** activations to introduce non-linearity.
- **Max Pooling** is selectively applied to downsample feature maps and encourage the network to retain only the most informative features.
- An **Adaptive Average Pooling** layer before the fully connected layers, allowing for flexible input sizes.
- Pretrained on our non-corn dataset, creating a 35-class plant disease feature extractor (backbone), ensuring it learns rich, plant disease features without exposure to corn.

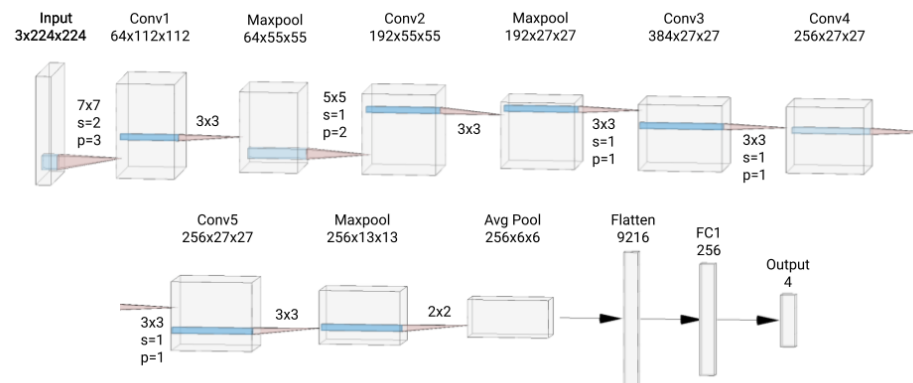


Figure 5: Layer-by-layer structure of LargeNet, showing convolutional, pooling, and fully connected stages.

3.2 CORN HEAD

For corn classification, we replace the 35-class classifier with a compact head with Flatten (9216) → Linear (256) → ReLU → Linear (4) producing logits for our 4 classes: Gray Leaf Spot, Healthy, Northern Leaf Blight, and Common Rust. The backbone is initialized from the pretrained 35-class weights, while the corn head is randomly initialized. A complete layer-by-layer schematic of the architecture is presented in Figure 5, detailing convolutional kernel sizes, strides, and pooling operations. This visual representation clearly illustrates the flow of data from input to output through the network.

3.3 TRAINING DETAILS

We conducted hyperparameter tuning on key hyperparameters including batch size, learning rate, and frozen layers. The final selected model used a batch size of 64, learning rate of 0.001, and **0 frozen layers** (the entire backbone was unfrozen and fine-tuned alongside the new corn head).

4 BASELINE MODEL

For the baseline model, the team used a simple Convolutional Neural Network (CNN) to classify corn leaf diseases. Since the task involves image data, a CNN was a reasonable starting point because it can pick up on patterns and textures directly from the raw images. The goal of this baseline was to have an easy-to-implement model, while still being strong enough to compare with the improved version later.

4.1 SIMPLE CONVOLUTIONAL NEURAL NETWORK

The architecture consisted of two convolutional layers, each followed by a ReLU activation and max pooling. The resulting feature maps were flattened and passed through a fully connected layer for final classification. We deliberately kept the design minimal—no batch normalization, dropout, or other regularization techniques were used—although heavy data augmentation was applied to improve generalization. A visual overview of the architecture is provided in Figure 6.

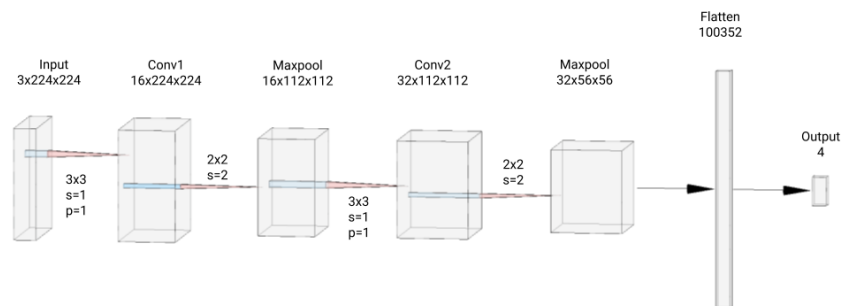


Figure 6: Baseline Model Architecture with labeled layer dimensions

Despite the model being simple, it still performed well. It reached a validation accuracy of 82.15% and a test accuracy of 86.81%, with corresponding losses of 0.4157 and 0.3351. The training and validation curves over 10 epochs are shown below in Figure 7 and give an overview of how the model improved over time.

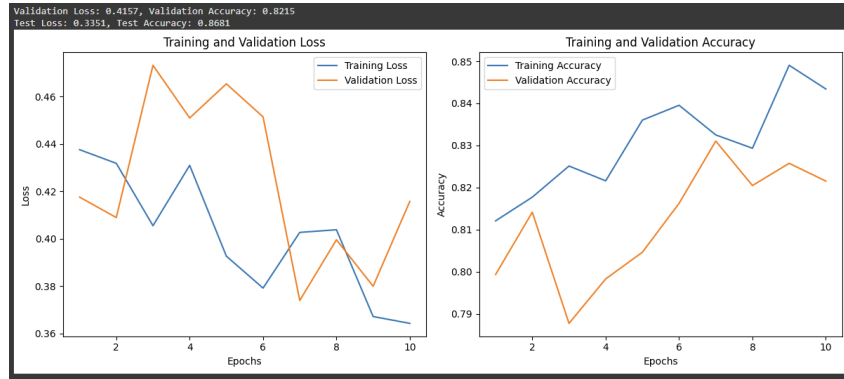


Figure 7: Training and validation curves over 10 epochs

The team also looked at some of the predictions made by the model to get a better idea of how it was doing qualitatively. Figure 8 shows test images along with their true and predicted labels. Green text means the model was correct, and red text means the prediction was wrong.

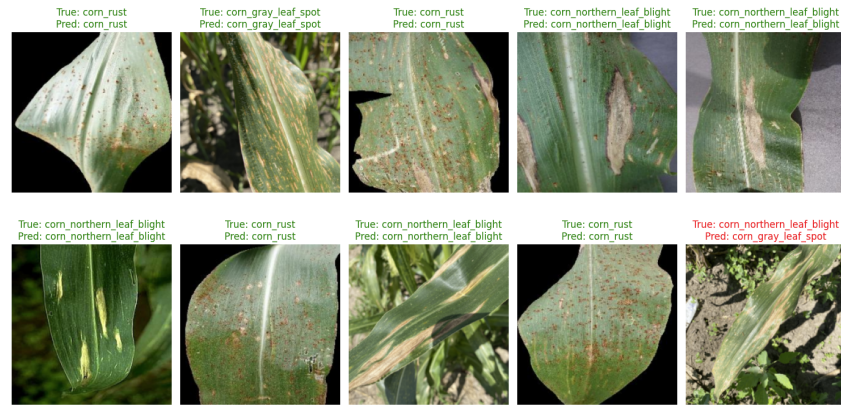


Figure 8: Model predictions. Green represents correct estimation, red represents wrong prediction.

From this, it is clear that the model was mostly correct on images that had clear disease patterns, like corn rust. However, it sometimes confused similar-looking diseases, especially northern leaf blight and gray leaf spot as they can look visually alike in real-life scenarios.

Overall, this simple CNN worked as a strong starting point. It gave a clear picture of what a basic model could achieve and helped set the foundation for improving the architecture later on.

5 QUANTITATIVE RESULTS

After experimenting with different configurations, the best-performing LargeNet setup was:

- Dropout: 0.5
- Batch Size: 64
- Learning Rate: 0.001
- Optimizer: Adam
- Epoch: 50

This configuration achieved a final test accuracy of 94.10%. The training curves for accuracy, error, and loss over the 41 training epochs are shown in Figure 9. The model's accuracy improved steadily

throughout training, with the training accuracy eventually reaching 99.64% and the validation accuracy plateauing around 94.5% and peaking at 94.79%.

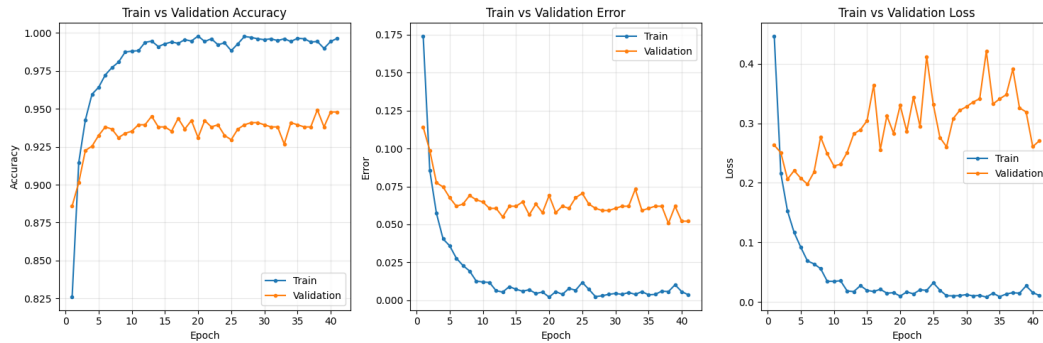


Figure 9: Training Curves for LargeNet

The error curves show a consistent decline for both training and validation, with final validation error rates converging to approximately 0.05–0.06, indicating strong classification performance. Similarly, the training and validation loss decreased significantly over the first few epochs and then stabilized at low values. In the later stages of training, a slight divergence emerged, training loss continued to decrease while validation loss began to rise, suggesting the onset of mild overfitting. However, this effect was small, as validation accuracy remained stable and high, indicating that LargeNet retained robust generalization to unseen corn images.

6 QUALITATIVE RESULTS

To better understand the performance of our LargeNet model, we examined its predictions on a subset of test images (Figure 10). The model demonstrates strong classification capabilities, correctly identifying most leaf diseases and healthy samples. Correct predictions are shown in green, while misclassifications are shown in red.

Overall, LargeNet performs well on clear, well-lit images where the disease symptoms are visually distinct, such as the rust pustules or the lesions of gray leaf spot. The correctly classified samples confirm that the model has successfully learned discriminative features for each corn disease category.

However, certain misclassifications reveal the model’s weaknesses. For example, gray leaf spot is sometimes confused with northern leaf blight due to the similarity in patterns, especially in images with overlapping symptoms or poor lighting. The model also shows a tendency to struggle in **in-field conditions**, likely because the majority of its training data came from the PlantVillage dataset, which is lab-curated with consistent lighting, centered leaves, and clean backgrounds. In contrast, real-world field images often contain cluttered backgrounds, variable zoom levels, and partially obstructed leaves, which can reduce classification accuracy.

Another challenge is **extreme lighting conditions**. Overly bright or harsh sunlight can wash out key features and alter the perceived color of lesions, causing the model to misinterpret disease patterns. This is evident in some of the false predictions where lesion boundaries appear faded or indistinct due to glare.

These qualitative observations align with the quantitative metrics, where the validation accuracy plateaus around 94–95%. While the model generalizes well to data similar to its training set, the few misclassifications highlight potential areas for improvement, such as incorporating more diverse, field-collected images and using targeted augmentations (e.g., brightness/contrast variation) to improve robustness.

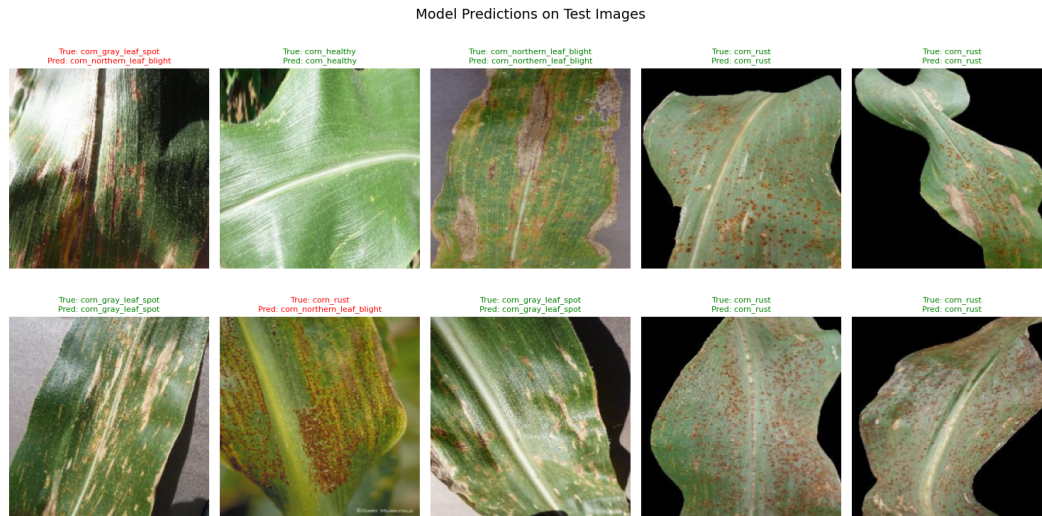


Figure 10: Sample LargeNet predictions on test images. True labels (top) and predicted labels (bottom) are shown, with correct predictions in green and incorrect in red.

7 EVALUATING THE MODEL ON NEW DATA

To ensure our results reflect the model’s performance beyond the training and validation distributions, we evaluated LargeNet on two completely unseen datasets containing leaf images from sources not used during training or hyperparameter tuning. Roboflow Corn Diseases comprises 3,647 labeled corn leaf images across Common Rust (1,177), Gray Leaf Spot (1,161), and Leaf Blight (1,309) captured under varied real-world lighting, angles, and backgrounds. IDADP Corn contains 3,600 images evenly split across 6 categories, from which we focused on Common Rust, Gray Leaf Spot, and Northern Corn Leaf Blight. None of these images were used in any stage of training or validation, ensuring a fair measure of generalization. All new test images went through the same resizing procedure (224×224 RGB & normalized) as the training data with no applied augmentations.

When tested on this new dataset, LargeNet achieved an accuracy of 96.44%, exceeding its validation accuracy of 94.79%. This improvement suggests that the model’s learned representations are robust and transferable to real-world corn disease classification, even when the data source changes.

However, deeper inspection revealed that the model’s strongest performance was on images resembling the PlantVillage style—clear, centered leaves with uniform lighting—while accuracy dropped slightly in challenging in-field conditions. These field images often featured cluttered backgrounds, varying zoom levels, partial occlusions, and harsh or uneven lighting, all of which sometimes caused misclassifications. A pattern we noticed in the misclassifications is that generally there were few false negatives; our model would detect the wrong disease instead of no disease. The confusion was between disease classes, not with healthy leaves. This emphasizes that the core challenge lies in differentiating visually similar diseases under challenging field conditions, not in incorrectly labeling healthy specimens.

To mitigate this gap, we propose two future improvements. The first is incorporating more diverse in-field samples into training to better capture real-world variability. Second, applying targeted data augmentation strategies such as brightness/contrast variation, random cropping, and background clutter simulation to improve robustness.

Overall, the strong performance on both familiar and unfamiliar datasets, combined with a clear understanding of its limitations, demonstrates that LargeNet is well-suited for deployment in practical corn disease detection scenarios, provided that it is trained further on more diverse image sources.

8 DISCUSSION

Our results demonstrate that the LargeNet model, pretrained on diverse non-crop datasets and fine-tuned on the CD&S dataset, achieved strong classification performance, with a test accuracy of 94.10% and minimal overfitting. This is a substantial improvement over our baseline CNN's 86.81% accuracy. The steady improvement in training and validation curves suggests that our training process effectively balanced bias and variance. Utilizing deeper architecture and tools like batch normalization, dropout, and adaptive pooling allows the model to extract more transferable features, even when excluding corn leaf images during pretraining.

When evaluated on entirely new, real-world corn leaf images not used during training, our model achieved 96.44% accuracy. This high performance on completely unseen data highlights the model's generalization ability and confirms that the learned features from pretraining transferred effectively to corn disease detection. This indicates that the model is usable in realistic agricultural scenarios, as it maintains reliability beyond the controlled test set.

An interesting aspect of our results is that the model generalized well despite the domain shift from pretraining datasets to our corn dataset. By excluding corn images during pretraining, we tested whether the network could learn plant disease features common among crop species. The strong performance indicates that certain disease patterns such as color changes, texture disruptions and unique shapes are common across various plants. This finding suggests broader potential for multi-crop disease detection systems trained on shared features.

Our qualitative analysis revealed specific failure modes. Misclassifications often occurred when symptoms were at an early stage or photographed under uneven lighting. Diseases with visually similar lesion patterns, such as Northern Leaf Blight and Gray Leaf Spot, were sometimes confused. These errors highlight the limitations of the current dataset's variability and suggest that expanding the training set to include more diverse real-world conditions could improve classification. Additionally, the model occasionally relied on background or texture cues, indicating that further regularization or background suppression targeting may be beneficial.

Through this project, we learned that model architecture and training strategies have a significant impact on domain transfer performance. Starting from a plant feature extractor allowed us to achieve high accuracy with limited corn-specific training data. We also confirmed the importance of balancing augmentation strength, where the model may overfit to specific conditions if too light and distort disease patterns if too heavy. Finally, this project reinforced the need for rigorous qualitative evaluations as numerical metrics alone may obscure meaningful error patterns that have real-world implications for farmer usage and reliability.

9 ETHICAL CONSIDERATIONS

A key ethical concern in deploying a corn disease classification model is the potential for users, such as farmers, to place excessive trust in the model's predictions without cross-verifying results. A false negative (failing to identify an actual disease) could result in serious crop loss and economic hardship. Conversely, a false positive may lead to the unnecessary use of pesticides, contributing to environmental harm and increased farming costs.

Another concern involves dataset bias. If the model is trained primarily on images collected under specific lighting conditions, geographic regions, or corn varieties, it may perform poorly when applied in different contexts. For example, the model might misclassify healthy corn as diseased when exposed to new image conditions, such as unfamiliar species or camera types, reducing its ability to generalize and fairness.

These considerations underscore the importance of transparent communication about model limitations, thorough validation across diverse conditions, and the continued involvement of human expertise in the decision-making process.

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